

LAE Strategic Partnerships & Business Models

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Low Altitude Economics

Week 6

Learning Objectives

By the end of this class, you should be able to:

- **Compare** asset-intensive, service-based, and platform-oriented LAE business models.
- **Explain** when a public-private partnership may be appropriate and what it does not solve.
- **Design** payment mechanisms and KPIs that align service quality, affordability, and commercial viability.
- **Allocate** demand, weather, operational, regulatory, and financing risks to the parties best able to manage them.
- **Evaluate** phased investment and contractual flexibility under uncertain market development.

Choosing the Commercial Architecture

Imagine two teams proposing the same LAE service.

- **Team A** wants to own aircraft, infrastructure, staff, and customer operations.
- **Team B** wants to coordinate specialist partners and sell an integrated service to the customer.

Neither model is automatically superior.

- Ownership can provide control but requires capital and operating capability.
- Coordination can provide flexibility but creates dependency, contracting, quality-control, and margin-sharing risks.
- **Quick poll (2 min):** Which capabilities should your Week 5 project own, contract, share, or buy as a service?

Session 1

Business Models & Partnership Structures

Concept Primer: Three Business-Model Archetypes

1. **Asset-intensive operator:** Owns or finances major aircraft and infrastructure and controls more of the operating system. High capital exposure, but potentially greater control and value capture.
2. **Service integrator:** Promises an outcome, such as delivery availability or inspection coverage, while leasing assets or contracting specialist operators. Lower asset ownership, but significant coordination and performance risk.
3. **Platform orchestrator:** Connects customers, operators, infrastructure, and data services. Lower physical-asset ownership, but meaningful software, cybersecurity, compliance, customer-acquisition, and network-development costs.

These are broad archetypes. Real businesses often combine elements of all three.

Applied Tool: Comparing Business Models

A Simple Monthly Economic-Profit Screen

$$\text{Monthly Economic Profit} = \text{Revenue} - \text{Cash OPEX} \\ - \text{Monthly Asset-Cost Allocation} - \text{Partner Payments}$$

- Asset ownership affects financing, residual value, utilization risk, and control.
- Contracting converts some investment into recurring partner payments, but does not eliminate liability or performance risk.
- Platforms may scale with relatively fewer physical assets, but require trust, liquidity on both sides, governance, data security, and sufficient transaction volume.
- Compare cash flow, economic cost, risk, control, and strategic capability rather than profit alone.

Back-of-the-Envelope: Comparing Three Archetypes

Assumption: Each model earns HK\$1 million monthly revenue. Figures are simplified and exclude tax and financing.

Asset-Intensive

- Cash OPEX: HK\$200k
- Asset allocation: HK\$333k
- Partner payments: HK\$0
- **Economic profit: HK\$467k**
- High capital and utilization exposure

Service Integrator

- Internal OPEX: HK\$100k
- Asset allocation: HK\$0
- Partner payments: HK\$800k
- **Economic profit: HK\$100k**
- Lower ownership, high supplier dependency

Platform

- Cash OPEX: HK\$150k
- Software allocation: HK\$33k
- Operator payouts: HK\$700k
- **Economic profit: HK\$117k**
- Network and governance risk

Takeaway: The ranking depends on partner payouts, utilization, demand, liability, and investment requirements. An asset-light label does not guarantee higher profit or lower risk.

Illustrative assumptions for classroom analysis; not market benchmarks.

Session 2

PPP Design, Contract Structures & Incentives

Concept Primer: Public-Private Partnerships

A PPP is a long-term arrangement in which a public authority and private party allocate responsibilities and risks for delivering infrastructure or services.

- A PPP is not simply government and business sharing cost and profit.
- The private party may design, finance, build, operate, maintain, or integrate specified components.
- The public authority defines outcomes, procurement rules, affordability, safety, access, performance, and accountability requirements.
- Payment may come from users, the public authority, or a combination.

Managerial test: Use a PPP only when long-term integration and risk allocation can create better value than conventional procurement or a normal service contract.

Applied Tool: Usage and Availability Payments

Usage-Based Component

$$R_U = p \times q$$

Availability-Based Component

$$R_A = A - D + u \times q$$

- A is the availability payment, D is deductions for unavailability or failed performance, and u is any usage payment.
- A usage-based model places more volume risk on the provider.
- An availability model can shift demand risk toward the public authority, but the provider retains performance risk through KPIs and deductions.
- Hybrid payment can balance readiness, activity, quality, and affordability.

Back-of-the-Envelope: How Payment Design Changes Risk

Assumptions: Monthly fixed cost = HK\$500k; variable cost = HK\$200/mission; actual missions = 300.

Usage-Only Payment

- Fee: HK\$1,000/mission
- Revenue: HK\$300k
- Variable cost: HK\$60k
- Fixed cost: HK\$500k
- **Operating loss: HK\$260k**

Availability Plus Usage

- Availability payment: HK\$650k
- Usage payment: HK\$200/mission
- Total revenue: HK\$710k
- Total cost: HK\$560k
- **Operating profit before deductions: HK\$150k**

Takeaway: Availability payment stabilizes revenue but should depend on readiness and performance. Weather relief, deductions, and service obligations must be defined explicitly.

Illustrative assumptions; excludes insurance, tax, financing, and penalties.

Concept Primer: Contractual Flexibility and Real Options

A real option is the managerial flexibility to change an investment decision as new information arrives.

- Examples include delaying investment, staging construction, expanding after demand is proven, switching technology, or abandoning a project.
- A contract can preserve this flexibility through milestone gates, extension rights, break clauses, change procedures, or phased commitments.
- Not every flexible clause is freely exercisable. Trigger conditions, notice, compensation, approval, and termination consequences must be specified.
- Regulatory change is usually addressed through carefully drafted change-in-law and relief provisions, not a vague right to pause whenever rules change.

Applied Tool: A Simple Flexibility Screen

Decision-Tree Comparison

Expected Project Value = $\sum_s \text{Pr}(s) \times \text{Payoff under the chosen action in state } s$

- Compare a rigid commitment with a staged or contingent decision.
- Flexibility creates value only if management can observe new information and respond meaningfully.
- Waiting may also have costs: option fees, delay, lost market position, inflation, or expiry of approvals.
- NPV is not inherently flawed. A conventional NPV can incorporate planned decisions; real-options analysis makes flexibility more explicit.

Back-of-the-Envelope: The Value of Staging

Scenario: A full commitment costs HK\$50M and generates HK\$100M if conditions are favourable, otherwise zero. Each state has probability 50%.

Commit Now

- Favourable net payoff: HK\$50M
- Unfavourable net payoff: -HK\$50M
- Expected value: $0.5(50) + 0.5(-50)$
- **HK\$0M**

Pay HK\$5M to Wait

- Favourable: invest later, net payoff HK\$45M
- Unfavourable: do not invest, net payoff -HK\$5M
- Expected value: $0.5(45) + 0.5(-5)$
- **HK\$20M**

Takeaway: Staging has value in this simplified example because the decision occurs after uncertainty is resolved. Timing, discounting, delay cost, and whether information truly resolves must also be modelled.

Illustrative decision tree; not a market valuation.

Session 3

Global Benchmarks, Co-opetition & Sandbox X

Concept Primer: Co-opetition

Co-opetition means cooperating in selected activities while continuing to compete in others.

- **Possible cooperation:** shared vertiports, charging, communications, technical interfaces, safety data, and common operational standards.
- **Continued competition:** routes, price, reliability, service design, customer relationships, and specialized capabilities.
- **Potential benefit:** lower duplication, larger compatible networks, and faster market development.
- **Potential risk:** discriminatory access, information leakage, disputes over investment, or weakening of competition.

Managerial requirement: Define access, cost sharing, data governance, performance, liability, and exit arrangements.

Applied Tool: Sharing Facility Cost

Facility Cost per Movement

$$\text{Shared Facility Cost per Movement} = \frac{\text{Fixed Facility Cost}}{\text{Total Movements}} + \text{Variable Facility Cost}$$

- Greater total use can spread fixed facility cost over more movements.
- This is the facility's cost, not automatically each operator's charge.
- The contract must decide whether operators pay equally, by usage, by peak capacity, or by investment contribution.
- Sharing may reduce cost only if additional traffic does not create excessive congestion or expansion cost.

Back-of-the-Envelope: Facility Cost Before and After Sharing

Assumptions: Fixed facility cost = HK\$40,000/day and variable facility cost = HK\$100/movement.

50 Movements per Day

- Fixed cost per movement: HK\$800
- Variable cost: HK\$100
- **Facility cost:**
HK\$900/movement

100 Shared Movements per Day

- Fixed cost per movement: HK\$400
- Variable cost: HK\$100
- **Facility cost:**
HK\$500/movement
- Reduction: about 44%

Takeaway: Sharing reduces allocated facility cost under these assumptions. Operator ticket cost, pricing, and profitability require additional aircraft, staff, load-factor, and contract data.

Illustrative assumptions; movements are not seats or passenger tickets.

Different Institutional Approaches

- **Hong Kong, Sandbox X:** Controlled trials are used to test project concepts, operations, infrastructure, emergency response, and cross-boundary procedures.
- **European U-space:** A regulatory and digital-service framework for managing unmanned-aircraft operations in designated airspace.
- **United States:** FAA and interagency plans coordinate certification, operations, airspace, infrastructure, environment, community engagement, and longer-term AAM development.
- **GBA collaboration:** May combine manufacturing, operating, testing, finance, logistics, and professional-service capabilities across different jurisdictions.

These are not simple rankings of speed or flexibility. Each approach creates different requirements, evidence needs, partnership opportunities, and market-entry risks.

In-Class Exercise: Partnership Term Sheet

Scenario: A public authority seeks a drone service for municipal bridge inspections.

- **Format:** Government and operator teams in groups of 4–5
- **Draft a five-point term sheet:**
 1. scope, service period, and responsibilities;
 2. availability, usage, or hybrid payment mechanism;
 3. two measurable KPIs and associated deductions or incentives;
 4. allocation of weather, demand, regulatory, and performance risks; and
 5. one phased-expansion, change, or termination provision.
- **Timing:** 15 minutes negotiation plus 5-minute discussion.

Exercise Template: Negotiation Matrix

Issue	Public Authority	Operator
Service outcome	Reliable, affordable inspection	Deliverable, measurable scope
Payment	Value for money and budget control	Bankable and predictable cash flow
Performance	Strong KPIs and remedies	Controllable targets and fair relief
Risk	Transfer where value is improved	Accept only manageable, priced risk
Flexibility	Ability to adapt public service	Compensation for material changes

Principle: Allocate each risk to the party best able to control, absorb, mitigate, or insure it, while considering the price of transferring that risk.

Key Takeaways

- Asset ownership, service integration, and platform coordination involve different combinations of control, capital, capability, and risk.
- A PPP is justified by value for money and effective long-term risk allocation, not merely by a desire for public funding.
- Availability payments should be linked to measurable readiness and performance, with appropriate deductions and relief rules.
- Risks should be allocated to the party best able to manage them, not automatically transferred to either government or operator.
- Staged investment and contractual flexibility can create value, but waiting and adjustment also carry costs.
- Shared infrastructure can lower unit cost, provided access, governance, congestion, data, liability, and competition are managed carefully.

Week 6 Assignment Briefing

Commercial Model and Partnership Memo (maximum 1,200 words)

- **Due:** Beginning of Week 7 (group assignment)
- **Task:** Design a commercially and institutionally credible partnership for a Hong Kong or GBA use case.
- **Requirements:**
 1. Recommend an asset-intensive, service-integrator, platform, or hybrid business model.
 2. Explain why a PPP, conventional procurement, or private contract is most appropriate.
 3. Design an availability, usage, or hybrid payment mechanism.
 4. Specify two measurable KPIs with deductions, incentives, and relief conditions.
 5. Allocate at least four major risks and justify each allocation.
 6. Draft one phased-investment, change, or termination provision and explain its limitation.
- **Grading focus:** Economic reasoning (40%), credible partnership design (30%), risk allocation (20%), professionalism (10%).

LAE Policy and Regulation

Preview question for next week:

How will government regulatory instruments—like noise curfews and BVLOS flight rules—completely rewrite the contracts we designed today?

Thank You + Q&A

Pre-Class Readings Reminder (for Week 7):

- *McKinsey: Perspectives on AAM (Platform orchestration)*
- *AiRMOUR Guidebook (Partnership process guidance)*

Questions?

(Final 15 minutes reserved for extended Q&A)